A. Course Handout (Version 1.0)

Institute/School Name	Chitkara University Institute of Engineering and Technology		
Department Name	Department of Computer Science & Engineering		
Programme Name	Bachelor of Engineering (B.E.), Computer Science & Engineering		
Course Name	Back End Engineering-II	Session	2025-2026
Course Code	23CS008	Semester/Batch	5th/2023
L-T-P (Per Week)	0-0-8	Course Credits	04
Pre-requisite	Basic concepts of system design and Databases	NHEQF Level	5.5
Course Coordinator	Dr. SALIL BHARANY	SDG Number	4,8,9

CLO01	Implement the concept of Web Stack using Nodejs and JSON
CLO02	Examine how to Handle the server-side Application using EJS.
CLO03	Apply acquired knowledge and skills through practical projects and case studies, simulating real-world scenarios to reinforce learning and problem-solving abilities.
CLO04	Gain hands-on experience with deploying backend applications to various environments (e.g., cloud platforms, virtual machines) and understanding basic DevOps principles.

1. Objectives of the Course

The scope of the course is to Gain a comprehensive understanding of backend development concepts, including server-side languages, databases, APIs, and server management.

- Students will master one or more programming, such as Javascript and Node.js, focusing on their application in server-side operations.
- Develop skills in designing and implementing databases.
- Learn how to design, build, and test RESTful APIs to facilitate communication between front-end and back-end systems.

2. Course Learning Outcomes

After completion of the course, a student should be able to:

Course Learning Outcome (CLO)	POs*	CL*	KC*	Sessions
CLO01 – Implement the concept of Web Stack using Node.js and JSON.	PO1, PO2, PO3, PO5, PO9	K2	Factual, Procedural	**36**
CLO02 – Examine how to handle the server-side Application using EJS.	PO1, PO2, PO3, PO9, PO11	K2	Conceptual, Procedural	**12**
CLO03 – Apply acquired knowledge and skills through practical projects and case studies, simulating real-world scenarios.	PO1, PO2, PO3, PO5	K3	Conceptual, Procedural, Metacognitive	**18**
CLO04 – Gain hands-on experience with deploying backend applications and understanding basic DevOps principles.	PO3, PO4, PO5, PO9, PO11	K4	Conceptual, Procedural, Metacognitive	**24**
Total Contact Hours				**90**

Revised Bloom's Taxonomy Terminology
* PO's available at (shorturl.at/cryzF)
**Cognitive Level =CL
***Knowledge Categories = KC

Course Learning Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CLO01	H	H	M		H				H			
CLO02	H	H	H		H				H	M		
CLO03	M	H	H		H							
CLO04			H	M	H						H	

H=High, M=Medium, L=Low

3. ERISE Grid Mapping

Feature Enablement	Level (1-5, 5 being highest)
Entrepreneurship	1
Research	3
Innovation	2
Skills	5
Employability	4

4. Recommended Books:

Text Books:

B01 Beginning Node.js, Express & MongoDB Development, Greg Lim, 2nd edition, sept. 2020.

B02 Full Stack Javascript: Learn Backbone.Js Node.Js and Mongodb, Azat Mardon, 2nd edition, Apress, Jan.2019.

B03 Aws The Ultimate Step-by-Step Guide From Beginners to Advanced, Dabha Ltd., Oct. 2020.

B04 C. J. Date, A. Kannan and S. Swamynathan, *An Introduction to Database Systems*, Pearson Education, Eighth Edition, 2009.

B05 SQL in a Nutshell, 4th Edition by Kevin Kline, Regina O. Obe, Leo S. Hsu Released June 2022 Publisher(s):O'Reilly Media, Inc. ISBN: 9781492088868.

B06 *Designing Data-Intensive Applications*, 1st Edition by Martin Kleppmann, Released March 2017, Publisher(s): O'Reilly Media, Inc., ISBN: 9781449373320.

B07 *Testing JavaScript Applications*, 1st Edition by Lucas Fernandes da Costa, Released May 2021, Publisher(s): Manning Publications, ISBN: 9781617297915.

Reference Books:

B05: SQL in a Nutshell, 4th Edition by Kevin Kline, Regina O. Obe, Leo S. Hsu Released June 2022

Publisher(s): O'Reilly Media, Inc. ISBN: 9781492088868.

E-Resources:

- https://library.chitkara.edu.in/subscribed-books.php

5. Other readings and relevant websites:

Serial No	Link of Journals, Magazines, websites and Research Papers
1.	https://byjus.com/gate/difference-between-sql-and-nosql/
2.	https://www.youtube.com/watch?v=SBY-YlHRTZ0
3.	https://www.w3schools.com/nodejs/
4.	https://nodejs.dev/learn
5.	https://youtu.be/c2M-rlkkT5o?feature=shared
6.	https://www.youtube.com/watch?v=Z3SYDTMP3ME
7.	https://www.youtube.com/watch?v=c2M-rlkkT5o
8.	https://www.youtube.com/watch?v=Z3SYDTMP3ME

6. Recommended Tools and Platforms

Code Quotient, SQL, NO-SQL, MongoDB

7. Course Plan:

Lecture Code	Topic	Sub-Topic	Chapter(s)
1	Introduction to Node.js	What is Node.js	B01 – Chapter 1
2		Why use Node.js	B01 – Chapter 1
3		Node.js vs Browser	B01 – Chapter 1
4		Node.js Architecture	B01 – Chapter 1
5		Running Node.js Code (CLI + IDE)	B01 – Chapter 1
6	Modules	Intro to Modules + require()	B02 – Chapter 2
7		CommonJS (Exports, Imports)	B02 – Chapter 2
8		ESM Modules Syntax and Use	B02 – Chapter 2
9		Module Resolution and Scope	B02 – Chapter 2
10		Creating Custom Modules	B02 – Chapter 2
11	Package Managers	Introduction to npm	B02 – Chapter 2
12		Creating Packages with npm	B02 – Chapter 2
13		Installing External Packages	B02 – Chapter 2
14		Understanding package.json	B02 – Chapter 2
FA-1			
15	Error Handling	Intro to Error Handling	B02 – Chapter 3
16		Uncaught Exceptions	B02 – Chapter 3

17		Handling Errors in Callbacks	B02 – Chapter 3
18		Handling Async/Await Errors	B02 – Chapter 3
19		Call Stack & Stack Trace	B02 – Chapter 3
20		Debugging with console & Node Inspect	B02 – Chapter 3
21	Starting Node.js Apps	Understanding Async Programming	B01 – Chapter 10
22		Event Loop and Callbacks	B01 – Chapter 10
23		Promises and Chaining	B01 – Chapter 10
24		Async/Await Deep Dive	B01 – Chapter 10
25		Reading/Writing Files Sync & Async	B02 – Chapter 3
26		File System Operations in CLI Tools	B02 – Chapter 3
27		CLI App Arguments using process.argv	B02 – Chapter 3
28		Building a CLI Project	B02 – Chapter 3
FA-2			
29	Working with APIs	Consuming External APIs with http/axios	B01 – Chapter 10
30		Designing Your Own API	B01 – Chapter 11
31		Handling API Responses	B01 – Chapter 11
32		Keeping App Running: PM2 Basics	B02 – Chapter 3
33		Nodemon & Dev Workflows	B02 – Chapter 3
34	Templating Engines	Why Templating + Intro to EJS	B01 – Chapter 17, B02 – 18
35		EJS Syntax and Placeholders	B01 – Chapter 17, B02 – 18
36		EJS Loops, Conditionals	B01 – Chapter 17, B02 – 18
CE-1 Syllabus (Lecture number 1-36)			
37		Layouts and Partials in EJS	B01 – Chapter 17, B02 – 18
38	Express.js	Express.js Setup and Basic Route	B01 – Chapter 6
39		Route Parameters and Query Strings	B01 – Chapter 7
40		REST API Design (GET, POST)	B01 – Chapter 7
41		REST API (PUT, DELETE)	B01 – Chapter 8
42		Express Middleware – Concepts	B02 – Chapter 6
43		Custom Middleware + Usage	B02 – Chapter 6

44		Express Error Middleware	B02 – Chapter 6
45		EJS with Express	B02 – Chapter 18
46	Express Security	Auth vs AuthZ Concepts	B01 – Chapter 10
47		Token Based Authentication (Overview)	B01 – Chapter 11
48		Protecting Routes	B01 – Chapter 11
FA-3			
49	Express WebSocket	WebSocket Concepts	B02 – Chapter 13
50		WebSocket Integration with Express	B02 – Chapter 13
51	Express + DB	MongoDB Driver Intro	B02 – Chapter 9
52		Connecting MongoDB with Express.js	B02 – Chapter 10
53	Git	Git Basics – init, add, commit	Git Docs
54		Git Branching	Git Docs
55	GitHub	GitHub Workflow, Push & Pull	Git Docs
56		Git Collaboration (Fork, PRs)	Git Docs
57	Relational DBs	PostgreSQL Setup	B05 – Chapter 4
58		PostgreSQL Basic Queries	B04 – Chapter 9
59		PostgreSQL Joins	B04 – Chapter 11
60		PostgreSQL Constraints	B04 – Chapter 13
61	MariaDB	MariaDB Intro + Setup	B05 – Chapter 3
62		MariaDB Table & Query Practice	B02 – Chapter 6
63		MariaDB Views and Stored Procedures	B02 – Chapters 9–10
FA-4			
64	NoSQL Databases	MongoDB Document Model	B06 – Chapter 2
65		MongoDB CRUD	B06 – Chapter 2
66	Time Series DB	InfluxDB – Concept & Use Cases	B06 – Chapter 2
67	Graph DB	Neo4j – Concepts and Use	B06 – Chapter 2
68	Key-Value Store	Redis Intro	B06 – Chapters 2–3
69		Redis – Data Structures	B06 – Chapters 2–3
CE-2(Lecture number 37-69)			
70	Scaling DBs	DB Indexing	B06 – Chapter 3
71		Sharding Concepts	B05 – Chapter 2
72		Sharding Strategies in Practice	B05 – Chapters 3–4
73		Replication Basics	B06 – Chapter 6
74		Replication Types	B06 – Chapter 6
75	Caching	Browser-side Caching	B06 – Chapter 5
76		CDN Concepts	B06 – Chapter 11

77		Redis for Server-Side Caching	B06 – Chapter 11
78	Web Security	Hashing – MD5/SHA Concepts	B01 – Chapter 11
79		HTTPS, SSL/TLS Setup	B02 – Chapter 12
80		Certificates & Encryption Summary	B02 – Chapter 12
81	Testing	Unit Testing	B07 – Chapters 1, 2
82		Functional Testing	B07 – Chapter 3
83		Integration Testing	B07 – Chapter 4
84	Deployment	Introduction to AWS & IAM	B03 – Chapter 1
85		AWS – EC2	B03 – Chapter 2
86		AWS – RDS	B03 – Chapter 3
87		AWS – Route 53	B03 – Chapter 4
88		AWS – S3	B03 – Chapter 5
89		Docker	B03 – Chapter 6
90		AWS Elastic Beanstalk	B03 – Chapter 7
ETE Syllabus (Lecture number 1-90)			

8. Delivery/Instructional Resources

Lecture	Topics	Web References	Audio-Video
1–5	Introduction to Node.js	https://nodejs.org/en/abouthttps://nodejs.dev/en/learn	https://www.youtube.com/watch?v=ENrzD9HAZK4
6–10	Modules (CommonJS, ESM, Custom Modules)	https://nodejs.org/api/modules.htmlhttps://blog.logrocket.com/understanding-es-modules-node-js/	https://www.youtube.com/watch?v=JVMZAfA2G6Mhttps://www.youtube.com/watch?v=d3x4ZcAqbYk
11–14	npm & Package Management	https://docs.npmjs.com https://nodesource.com/blog/the-basics-of-package-json/	https://www.youtube.com/watch?v=jHDhaSSKmB0
15–20	Error Handling & Debugging	https://nodejs.org/api/errors.htmlhttps://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide/Control_flow_and_error_handling	https://www.youtube.com/watch?v=H4VZ3UYZJgYhttps://www.youtube.com/watch?v=Nt-qa_LlUH0
21–28	Async Programming & CLI Apps	https://developer.mozilla.org/en-US/docs/Learn/JavaScript/Asynchronoushttps://nodejs.dev/en/learn	https://www.youtube.com/watch?v=PoRJizFvM7s
29–33	Working with APIs	https://axios-http.com/docs/introhttps://nodejs.org/api/http.html	https://www.youtube.com/watch?v=8zKuNo4ay8Ehttps://www.youtube.com/watch?v=GZvSYJDk-us
34–37	Templating with EJS	https://ejs.cohttps://expressjs.com/en/guide/using-template-engines.html	https://www.youtube.com/watch?v=BUKz6N5eOto

38–45	Express.js Core Features	https://expressjs.com/en/starter/hello-world.htmlhttps://expressjs.com/en/guide/routing.html	https://www.youtube.com/watch?v=L72fhGm1tfEhttps://www.youtube.com/watch?v=SccSCuHhOw0
46–48	Express Security (JWT, Auth)	https://jwt.io/introductionhttps://auth0.com/learn/json-web-tokens/	https://www.youtube.com/watch?v=2jqok-WgelI
49–50	WebSocket with Express	https://socket.io/docs/v4/https://developer.mozilla.org/en-US/docs/Web/API/WebSockets_API	https://www.youtube.com/watch?v=rxzOqP9YwmM
51–52	MongoDB + Express Integration	https://www.mongodb.com/docs/drivers/node/https://www.mongodb.com/developer/languages/javascript/quickstart/	https://www.youtube.com/watch?v=fgTGADljAeg
53–54	Git Basics & Branching	https://git-scm.com/docshttps://www.atlassian.com/git/tutorials/using-branches	https://www.youtube.com/watch?v=USjZcfj8yxE
55–56	GitHub Workflow	https://docs.github.com/en/get-started/quickstarthttps://opensource.guide/how-to-contribute/	https://www.youtube.com/watch?v=RGOj5yH7evk
57–60	PostgreSQL Setup & Joins	https://www.postgresqltutorial.comhttps://www.geeksforgeeks.org/sql-join-set-1-inner-left-right-and-full-joins/	https://www.youtube.com/watch?v=qw--VYLpxG4
61–63	MariaDB Tables & Procedures	https://mariadb.com/kb/en/library/sql-statements/https://mariadb.com/kb/en/stored-procedures-and-functions/	https://www.youtube.com/watch?v=cN5gCEpV-CM
64–65	MongoDB NoSQL CRUD	https://www.mongodb.com/docs/manual/crud/https://www.mongodb.com/docs/manual/core/data-models/	https://www.youtube.com/watch?v=excAOvwF_Wk
66	Time Series DB – InfluxDB	https://docs.influxdata.com/influxdb/latest/https://www.influxdata.com/time-series-database/	https://www.youtube.com/watch?v=V2HDhju6vOc
67	Graph DB – Neo4j	https://neo4j.com/developer/get-started/https://www.tutorialspoint.com/neo4j/index.htm	https://www.youtube.com/watch?v=8jNPelugHaw
68–69	Redis – Key-Value Store & Structures	https://redis.io/docs/getting-	https://www.youtube.com/watch?v=E6urxv6X6T0

		started/https://redis.io/docs/data-types/	
70–74	Scaling DBs: Indexing, Sharding, Replication	https://docs.mongodb.com/manual/sharding/https://use-the-index-luke.com/	https://www.youtube.com/watch?v=UjNjeQKqX18https://www.youtube.com/watch?v=77gLZAfK6p4
75–77	Caching: Browser, CDN, Redis	https://developer.mozilla.org/en-US/docs/Web/HTTP/Cachinghttps://redis.io/docs/manual/client-side-caching/	https://www.youtube.com/watch?v=al3ch7aY2JA
78–80	Web Security – Hashing, SSL, TLS	https://www.cloudflare.com/learning/ssl/what-is-ssl/https://en.wikipedia.org/wiki/Cryptographic_hash_function	https://www.youtube.com/watch?v=SJJmoDZ3il8
81–83	Testing – Unit, Functional, Integration	https://testing-library.com/docs/https://jestjs.io/docs/getting-started	https://www.youtube.com/watch?v=r9HdJ8P6GQI
84–90	Deployment – AWS, Docker, Elastic Beanstalk	https://docs.aws.amazon.com/https://docs.docker.com/get-started/	https://www.youtube.com/watch?v=ulprqHHWlnghttps://www.youtube.com/watch?v=V7x9SKZ8LtE

9. LAB COMPONENT

Lab No.	Lab Title	Mapped Lecture Range	Resources
Lab 1	Set up a Node.js Project with Core Modules	1–20	Node.js Tutorial for Beginners — Node.js basics, modules, npm, error handling
Lab 2	Build a CLI-Based Task Manager	21–32	Build a CLI App in Node.js — process.argv, CLI inputs, file handling, async/await
Lab 3	Create a Dynamic Website using EJS & Express	33–40	EJS Templating in Express — EJS basics, layout, partials, form handling
Lab 4	Develop RESTful APIs with Express Middleware	41–48	Build a REST API with Express — CRUD operations, Express Router, middleware, JWT overview
Lab 5	Build a Chat App with WebSocket and MongoDB	49–56	WebSocket Chat App with Node.js + MongoDB & Express
Lab 6	Set up Relational Databases: PostgreSQL & MariaDB	57–63	PostgreSQL Tutorial + MariaDB Beginners Guide
Lab 7	Create NoSQL Applications with MongoDB & Neo4j	64–67	MongoDB Full CRUD + Intro to Neo4j + InfluxDB intro
Lab 8	Implement Redis Caching and DB Scaling	68–74	Redis Crash Course + Database Indexing, Sharding
Lab 9	Secure Apps with Caching, Hashing & SSL/TLS	75–80	Node.js Authentication Security + SSL/TLS Certificate Setup
Lab 10	Deploy a Node.js App on AWS using Docker	81–90	Deploy Node.js on AWS EC2 + Dockerize Node App + AWS Beanstalk

10. Action plan for different types of learners

Slow Learners	Average Learners	Fast Learners
• Remedial Classes on Saturdays • Encouragement for improvement using Peer Tutoring • Use of Audio and Visual Materials • Use of Real-Life Examples	• Workshops • Formative Exercises used to highlight concepts and notions • E-notes and E-exercises to read ahead of the pedagogic material.	• Engaging students to hold the hands of slow learners by creating a Peer Tutoring Group • Design solutions for complex problems • Design solutions for complex problems • Presentation on topics beyond those covered in CHO

11. Evaluation Scheme & Components:

Evaluation Component	Type of Component	No. of Assessments	Weightage of Component	Mode of Assessment
Component1	Formative Assessment	4*	20%	offline
Component 2	Continuous Evaluation (CE)	02**	30%	Offline
Component 3	End Term Examinations (Project Viva)	01**	50%	offline
Total		**100%**		

*** Average of best 3 Formative Assessments will be counted for final assessment.**
**** Student needs to appear for both continuous evaluations and average will be counted for final assessment.**
**As per Academic Guidelines, a minimum of 75% attendance is required to appear for the End Term Examination.

12. Syllabus of the Course:

Subject: Back End Engineering (22CS026)

S. No.	Topic(s)	No. of Sessions	Weightage %
1	Introduction to Node.js including its comparison with the browser and how to run Node.js code, exploring modules such as CommonJS and ESM, and learning how to create custom modules. Working with npm to create and install packages. Understanding error handling in Node.js including uncaught exceptions, handling asynchronous errors, analyzing the call stack and stack trace, and using debugging tools. Beginning asynchronous programming, file handling, command-line apps, interacting with APIs, keeping applications running, and using templating engines. Introduction to Express.js, building RESTful APIs, routing, middleware, error handling, and implementing EJS templating. This section covers foundational server-side concepts using Node.js and Express.js.	36	40%

CE-1 (Covers the first 40% of the syllabus)			
2	Deep dive into authentication and authorization in Express.js, JWT implementation, WebSocket integration, and connecting Express.js with MongoDB. Introduction to version control using Git and GitHub. Relational databases including PostgreSQL and MariaDB (I & II), followed by NoSQL databases such as MongoDB, InfluxDB, Neo4j, and Redis. Database scaling strategies including indexes, sharding, and replication. Covers client-side and server-side caching (Redis), and web security practices including MD5, SHA, HTTPS, and TLS.	33	40%
CE-2 (Covers the next 40% of the syllabus)			
3	Covers software testing techniques such as integration, unit, and functional testing. The final deployment module includes working with AWS IAM, EC2, RDS, Route 53, S3, Docker, and Elastic Beanstalk**.** This part ensures learners can test, deploy, and scale full-stack applications in production environments.	21	20%
End Term	*(Covers 100% of the syllabus)*		

This Document is approved by:

Designation	**Name**	**Signature**
Course Coordinator	Dr. Salil Bharany	
Head-Academic Delivery	Dr. Susheela Hooda	
Dean	Dr. Rupali Gill	
Date	24.06.2025	